

LIBEREC, Czech Republic

WMO: 116030

Lat: 50.7699N

Lon: 15.0240E

Elev: 1317

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	13.3	1.7	6.3	10.4	7.1	8.3	15.9	27.2	38.9	23.2	35.9	4.5	300	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	84.9	66.3	81.3	64.8	78.0	63.5	68.2	80.1	66.5	77.9	64.9	75.2	6.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	94.4	72.2	62.6	89.1	70.3	61.0	84.3	69.0	33.2	80.0	31.8	78.3	30.6	75.3	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.5	15.4	1.6	90.4	7.2	3.8	-3.6	93.1	-7.8	95.4	-11.8	97.5	-17.0	100.3
			1.1	71.8	7.2	1.6	-4.0	72.9	-8.2	73.8	-12.2	74.7	-17.4	75.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	29.4	31.8	37.8	46.9	54.6	60.3	64.0	63.5	55.8	47.8	39.2	31.9	(6)
(7)		DBStd	14.27	8.59	8.52	7.47	8.21	7.13	6.82	6.33	6.12	6.45	7.22	7.18	7.97	(7)
(8)		HDD50	2757	639	511	382	152	34	3	0	0	17	129	328	562	(8)
(9)		HDD65	6747	1104	931	844	543	331	175	98	103	284	533	774	1027	(9)
(10)		CDD50	1659	0	0	3	60	176	311	434	419	191	61	4	0	(10)
(11)		CDD65	172	0	0	0	1	8	32	67	57	7	0	0	0	(11)
(12)		CDH74	1747	0	0	0	7	83	347	710	539	61	0	0	0	(12)
(13)		CDH80	447	0	0	0	0	7	80	211	143	6	0	0	0	(13)
(14)	Wind	WSAvg	6.9	8.5	8.0	7.6	6.7	6.2	5.9	5.7	5.3	5.8	7.0	8.0	8.5	(14)
(15)	Precipitation	PrecAvg	37.2	2.9	2.5	2.8	1.9	3.1	3.7	4.8	4.3	3.0	2.4	2.8	2.9	(15)
(16)		PrecMax	50.0	6.1	4.6	6.9	4.6	5.9	8.8	11.3	13.1	7.0	5.2	6.2	5.1	(16)
(17)		PrecMin	24.3	0.2	0.2	1.3	0.1	1.1	1.4	1.1	0.8	1.1	0.6	0.0	1.1	(17)
(18)		PrecStd	5.3	1.5	1.2	1.3	0.9	1.4	1.5	2.5	2.7	1.6	1.3	1.5	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	52.2	62.6	74.7	80.7	86.7	90.6	89.9	79.7	69.7	57.7	49.6	(19)
(20)			MCWB	46.1	45.3	49.3	57.4	63.2	68.0	67.4	66.6	63.6	58.2	52.1	46.5	(20)
(21)		2%	DB	46.0	48.7	57.2	69.9	76.2	82.0	85.5	83.7	75.1	64.5	54.4	46.5	(21)
(22)			MCWB	42.9	44.0	46.9	54.5	60.8	65.4	66.5	65.3	61.5	56.1	50.1	43.9	(22)
(23)		5%	DB	43.0	45.6	52.8	65.7	72.0	78.0	81.7	79.7	71.2	61.1	51.7	44.4	(23)
(24)			MCWB	40.4	41.8	45.2	52.5	58.7	63.3	65.0	64.4	60.1	55.0	48.2	42.1	(24)
(25)		10%	DB	40.2	42.7	48.7	61.6	68.3	73.9	77.6	76.1	67.2	58.2	49.2	41.9	(25)
(26)			MCWB	38.1	39.4	43.4	50.6	56.5	61.7	63.9	63.0	58.5	53.6	46.2	40.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	47.7	51.1	58.5	65.8	70.7	70.4	69.3	65.8	60.2	53.5	46.9	(27)
(28)			MCD B	49.4	50.8	59.9	71.9	76.7	83.4	82.5	82.4	74.3	66.1	56.3	49.0	(28)
(29)		2%	WB	43.1	44.6	48.3	56.0	62.6	67.4	68.2	67.5	63.5	57.6	50.8	44.5	(29)
(30)			MCD B	45.2	47.3	55.2	67.6	72.9	77.7	80.8	79.1	71.7	62.5	53.8	46.2	(30)
(31)		5%	WB	40.5	42.2	46.1	53.6	60.1	64.9	66.5	65.7	61.4	55.8	48.6	42.2	(31)
(32)			MCD B	42.8	45.0	51.4	63.5	69.5	74.9	78.8	76.8	68.9	60.1	51.2	44.2	(32)
(33)		10%	WB	38.1	39.5	43.7	51.3	58.0	62.8	64.8	64.1	59.4	54.0	46.4	40.1	(33)
(34)			MCD B	39.9	42.5	48.3	60.2	66.4	72.2	75.4	73.8	65.8	57.9	48.9	41.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	10.5	13.6	18.0	18.7	18.9	19.2	19.6	16.3	12.2	8.1	7.4	(35)
(36)			MCDBR	9.4	12.9	21.4	24.8	24.0	25.1	26.4	24.8	22.1	16.2	11.1	8.3	(36)
(37)			MCWBR	7.8	9.3	12.7	12.8	11.8	11.8	11.2	10.7	11.1	9.6	8.0	7.1	(37)
(38)		5% WB	MCD BR	9.0	11.2	18.0	22.0	21.5	22.3	23.0	21.8	19.3	14.7	10.4	8.2	(38)
(39)			MCWBR	7.8	8.9	11.6	12.0	11.6	11.7	10.9	10.4	10.7	9.5	8.1	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.326	0.369	0.412	0.413	0.411	0.428	0.422	0.383	0.358	0.331	0.298	(40)	
(41)		taud	2.342	2.319	2.251	2.191	2.230	2.266	2.223	2.261	2.341	2.380	2.394	2.376	(41)	
(42)		Ebn at Noon	228	254	261	262	267	269	261	256	254	238	215	212	(42)	
(43)		Edh at Noon	21	27	35	42	42	41	42	39	32	26	20	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	251	485	803	1296	1568	1675	1583	1398	998	571	267	182	(44)	
(45)		RadStd	42	77	120	172	180	137	188	137	135	82	35	18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	-295	+119	+47	(46)
(47)	Regional (46 neighbors)	+0.76	N/A	N/A	N/A	+0.61	+0.59	N/A	N/A	+95	+34	(47)

Nomenclature: See separate page